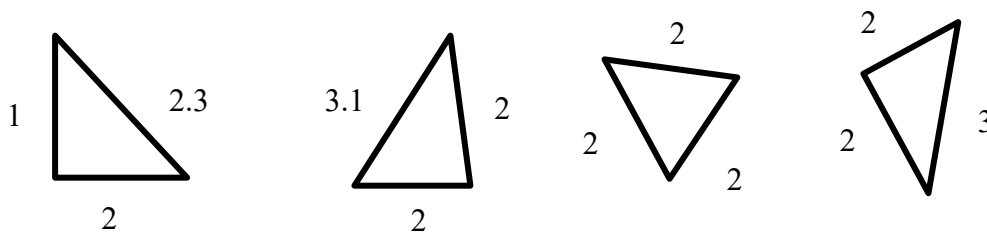


Week 3

Practical Exercises

Following from last week's exercise:

1. If there are multiple files in the directory where the source code is, modify the code so that the user gets to decide which file the programme will operate on.
 2. Consider what potential errors can occur in your code if the user did not use it correctly or if something went wrong and handle these errors in your code.
 3. Induces a situation where an error occurs inside a function, and explore how you can handle it.
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4. Starting from the code L5D1.cpp, modify the code such that after the user has input a sequence of numbers followed by a character, the programme calculates the mean, and the standard deviation of the input numbers and output the result back onto the command line. Implement functions to calculate these parameters.
 5. Convert the dynamic allocation code into a function for dynamic allocation that gets called whenever a new item is to be added to the array. Call the new function "push_back".
 6. Using the function "push_back", write a new code that creates and fills an array of 1000000 elements (fill it with any double of your choice). Use the "Chrono" library from Lecture 2 to time the execution of this code. How can you make the code run faster? (Hint: experiment with how the variable "size" is incremented).
 7. You have the following 4 triangles (not drawn to scale):



Write a programme that adds the perimeters of all the triangles, then adds their areas as well. The programme should do that by two methods:

- a. Using basic variables.
- b. Using classes and objects (use the code L6D1.cpp).

Which method did you find easier any why?